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Chair of measurement and Sensor System Techniques
Institute of Circuits and Systems

Student Research Project

Reservoir Computing for Multimodal Decomposition within Optical Networks

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Task for the preparation of a Student Research Project

Course:	M.Sc. Computational Modeling and Simulation
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Objectives of work

Task description: Multi-mode optical fibers are promised to enhance state-of-the-art single-mode networks in term of capacity by using their spatial domain for multiplexing. However,launched signals are distributed by modal scrambling during transmission within the fiber. Therefore it is necessary to measure the scrambling and calibrate the systems to deploy multiple spatial channels throughout the fibre.Although measurement can be implemented using digital holography, a second optical path is necessary for reference. This limitation can be overcome by using deep learning based on a intensity-only measurement for mode decomposition.Due to phase ambiguities a trade-off between prediction quality and response time has to be found. Within this work modal decomposition is implemented using reservoir computing. It is expected that using a optical reservoir the limitation of phase ambiguities is reduced.

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Statement of authorship

I hereby certify that I have authored this document entitled *Reservoir Computing for Multimodal Decomposition within Optical Networks* independently and without undue assistance from third parties. No other than the resources and references indicated in this document have been used. I have marked both literal and accordingly adopted quotations as such. There were no additional persons involved in the intellectual preparation of the present document. I am aware that violations of this declaration may lead to subsequent withdrawal of the academic degree.

Dresden, 9th May 2024

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Abstract

In the ever-evolving landscape of optical networking, the demand for increased capacity and efficiency continues to drive research towards innovative solutions.A promising candidate to resolve the capacity crunch of current singlemode networks is the use of multimode fibers. The increased core diameter allows the propagation of multiple parallel channels called modes. However, during propagation those orthogonal modes suffer from cross-talk, which can be characterized by complex digital signal processing or using digital holography. Another approach to compensate on cross-talk is given by a reference-less intensity-only measurment and the use of machine learning. The limiting factor here is the computational effort for a high number of decomposable channels and the phase ambiguity of optical fields when using intensity-only measurements. In this work, reservoir computing is investigated to address both, phase ambiguity and computational complexity of mode decomposition. The core concept of RC lies in harnessing the dynamics of a fixed, randomly initialized recurrent neural network, known as the reservoir, to transform input data through complex, nonlinear behavior into a high-dimensional space. By directing the output of the reservoir to a simple readout layer, typically a linear classifier, tasks such as classification and regression can be efficiently performed. Unlike traditional neural networks that learn all parameters, RC maintains a static reservoir while training only the readout layer, simplifying the training process and reducing computational complexity. In this research project, RC is implemented using an addidtional MMFs, the reservoir MMF with random mode coupling, where the inherent randomness and complexity of physical scattering processes within MMFs serve as the reservoir, enabling efficient computation at the speed of light. This unique approach allows MMFs to function as optical computing operators, enhancing the processing capabilities of optical systems.

In this project, mode decomposition is performed for a 5-mode fiber using a 55-mode MMF as reservoir and a intensity-only measurement. The readout-layer of the reservoir is optimized using 1k simulated samples of a randomly initialized MMF. After training, the NN performs mode decomposition with a correlation on reconstructed ground truth intensity images and a fidelity on the reconstructed fields. This is achieved without limiting the phase.This breakthrough underscores the potential of RC as a powerful computational framework for enhancing optical network performance and efficiency through innovative modal decomposition techniques. The modal decomposition using RC with multi-mode fibers in optical networks presents a promising avenue for future advancements in optical communication technologies, offering a novel approach to reference-less optical network charecterization through advanced optical technologies.

Contents

1 Introduction

In the realm of optical networking, the demand for increased capacity and efficiency continues to drive research towards innovative solutions.Multi-mode fibers (MMFs) are a promising candidate providing higher data rates and increased security. However, excitation of modes for such an MMFs leads to cross-talk at the transmitter side. To deploy orthogonal channels within the MMF, the cross-talk needs to be characterized. Therefore, the scrambled information needs to be decomposed into the excited modes, which is called mode decomposition. One such promising avenue is the integration of Reservoir Computing (RC) for multi-modal decomposition within optical networks. This research project delves into the application of RC to enhance the capabilities of optical networks, particularly focusing on multi-mode optical fibers (MMFs).

The core concept of Reservoir Computing lies in harnessing the dynamics of a fixed, randomly initialized recurrent neural network, known as the reservoir. By processing input data dynamically through complex, nonlinear behavior, the reservoir transforms the data into a high-dimensional space. Subsequently, the output of the reservoir is directed to a simple readout layer, typically a linear classifier, to perform tasks such as classification or regression. Unlike traditional neural networks, where all parameters are learned, RC maintains a static reservoir while training only the readout layer, simplifying the training process and reducing computational complexity.

Within the context of this research project, RC for mode decomposition is investigated using MMFs with random mode coupling. The inherent randomness and complexity of the physical scattering processes within MMFs serve as the reservoir, enabling efficient computation at the speed of light. This unique approach allows MMFs to function as optical computing operators, enhancing the processing capabilities of optical systems.

Overall, the utilization of Reservoir Computing for multi-modal decomposition within optical networks represents a significant advancement in the field, offering a novel approach to enhancing optical network capacity and efficiency through innovative computational frameworks and advanced optical technologies.

2 State of the Art and Scientific Question

2.1 Multimode Optical Fibre Communication

Fiber optics has emerged as the backbone of modern communication systems, revolutionizing the way data is transmitted over long distances. While wireless connections have gained popularity, especially for shorter distances, fiber optics remains indispensable for high-capacity, long-reach communication links. In critical environments such as data centers and intercontinental connections, where reliability and data integrity are paramount, fiber-based communication networks are the preferred choice. With the exponential growth of data transmission requirements driven by the digital revolution [1], the capacity of single-mode fibers, governed by Shannon's limit, has become a bottleneck. Single-mode fibers, despite their high bandwidth, face limitations in accommodating the ever-increasing data demands. This is where multimode fibers (MMFs) come into play.

A Multimode fiber (MMF) is an optical fiber designed to carry multiple light modes simultaneously [8] [10]. Unlike single-mode fibers that support only one mode of light propagation, MMFs allow various modes to propagate through the fiber core. MMFs are susceptible to modal dispersion, where different modes of light travel at different speeds, potentially limiting their performance over longer distances or at higher data rates. Therefore, losses increase with fibre length. MMFs are sometimes preferable for short-reach connections [7].

Mode dispersion is a significant issue in optical fiber communication systems, stemming from the fact that different modes of light propagate at varying speeds within the fiber[5]. While the propagation speed of individual modes would not be problematic if they did not overlap in time, mode dispersion also involves mode coupling. Mode coupling refers to the phenomenon where light from one mode of the fiber is transferred or coupled to another mode. When multiple modes with different propagation speeds and states interact within the fiber due to mode coupling [4], interference effects occur. This interference leads to constructive or destructive interference, where portions of light in the same state either reinforce each other (constructive) or cancel each other out (destructive). Throughout the length of the fiber, this superposition of multiple modes continuously changes in propagation direction and time, resulting in complex and varying optical characteristics along the fiber this is due to the disorder in the media and environment.Understanding and mitigating mode dispersion are crucial for optimizing the performance and reliability of optical communication systems. Effective strategies to address mode dispersion involve advanced fiber design, signal processing techniques, and modulation schemes tailored to minimize the impact of mode coupling and interference.

2.2 Machine Learning fundamentals and Concepts

Machine learning is a branch of artificial intelligence focused on developing algorithms and models that can learn from data and make predictions or decisions. Conventional modelling uses equations from the physics and engineering behind a problem.The fundamental concept of machine learning involves training a general model, like several matrix operations or filters to extract data from a data-set to recognize underlying patterns or relationships, which it can then apply to new, unseen data to generate predictions or classifications.

The machine learning process encompasses several essential steps. First, data collection involves gathering relevant data-sets and preparing them for training by cleaning, transforming and organizing the data into a suitable format for the learning algorithms. Next, model selection involves choosing an appropriate algorithm based on the problem type (e.g., classification, regression, clustering) and the characteristics of the data.

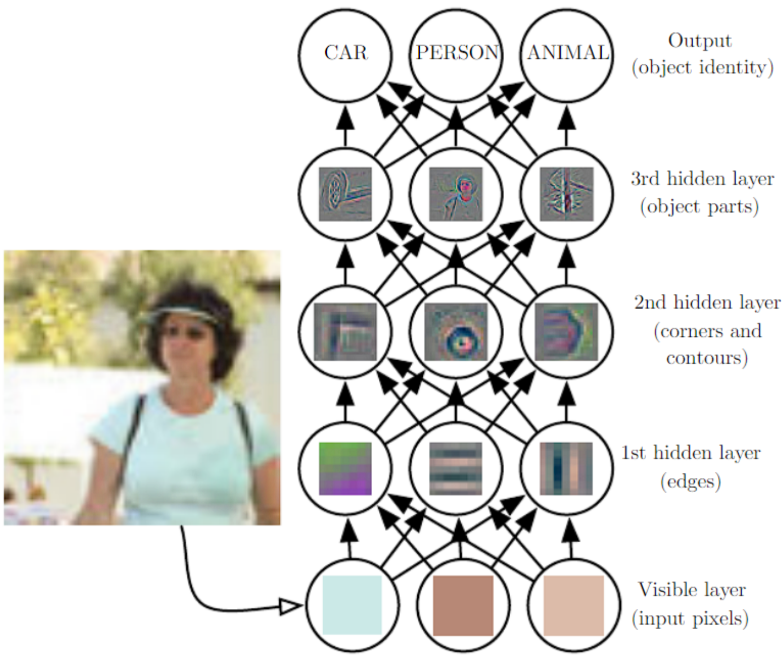

Figure 2.1: Illustration of a Deep Neural Network (DNN) model
[3]

Feature extraction and engineering are critical steps where informative features are identified and selected from the data to serve as inputs to the model. This involves transforming raw data into meaningful features that enhance the model's performance.However modern neural network models are capable of performing this task without human intervention. The model is then trained using a training data-set to optimize its parameters through an iterative process of adjusting weights or parameters to minimize errors between predicted outputs and actual targets.The weights are optimized by back-propagation, a batch of predicted images is then analyzed and the error compared to GT is used in the backwards path to optimize the weights.

Evaluation and validation are performed using separate validation data-sets to assess the model's performance and ensure it generalizes well to new data. In regression tasks, the Pearson correlation coefficient may be used to assess how well the predicted values correlate with the true values. Fidelity is often refers to how well a model preserves information or accurately represents the input data. It's commonly assessed using metrics like mean squared error (MSE) or cross-entropy loss. MSE measures the average squared difference between predicted and actual values.

Correlation: The correlation measures the linear relationship between two vectors x and y. It is calculated using the formula:

$$\text{Correlation} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

where x_i and y_i represent the individual elements of the vectors x and y respectively, and $\bar{x}$ and $\bar{y}$ represent their means.

Fidelity: The fidelity measures the similarity between two quantum states represented by vectors ψ_{output} and $\psi_{\text{ground truth}}$. It is calculated as the squared modulus of the inner product between the two vectors:

$$\text{Fidelity} = |\langle \psi_{\text{output}} | \psi_{\text{ground truth}} \rangle|^2$$

Hyper-parameters like learning rate, batch size are configuration settings that dictate the behavior and performance of a machine learning model. Unlike parameters, which are learned from the training data, hyper-parameters are set before the learning process begins and remain constant throughout training.Hyper-parameter tuning is conducted to fine-tune the model's parameters for optimal performance and prevent issues like over-fitting or under-fitting.

In terms of model architecture, machine learning models are typically composed of several layers that process input data sequentially to extract features and make predictions. Commonly used layers in neural networks include:

1. Input Layer: Receives input data and passes it to the subsequent layers.
2. Hidden Layers: Intermediate layers that perform transformations on the input data using weighted connections and activation functions.
3. Output Layer: Produces the final output or prediction based on the processed data from the hidden layers.

Once trained and validated, machine learning models can be deployed into production environments to make predictions. Continuous monitoring and updating of the model may be necessary to maintain its performance over time and adapt to evolving data patterns. Overall, machine learning is a powerful technology with diverse applications spanning image and speech recognition, natural language processing, healthcare diagnostics, and autonomous systems.

2.3 Reservoir Computing for Modal Decomposition

Reservoir Computing (RC) is a computational framework that harnesses the dynamics of a fixed, randomly initialized recurrent neural network (RNN) known as the reservoir[9] [6].The core idea behind RC is to exploit the intrinsic computational power of the reservoir, which consists of interconnected nodes (neurons) with random connections. The reservoir processes input data dynamically, and its complex, nonlinear behavior transforms the input into a high-dimensional space. The output of the reservoir is then fed into a simple readout layer (typically a linear classifier) to perform the desired task (e.g., classification or regression). Unlike traditional neural networks where all parameters are learned, in RC, only the readout layer is trained while the reservoir remains static. This setup simplifies training and reduces computational complexity, making it particularly attractive for large-scale data processing tasks.

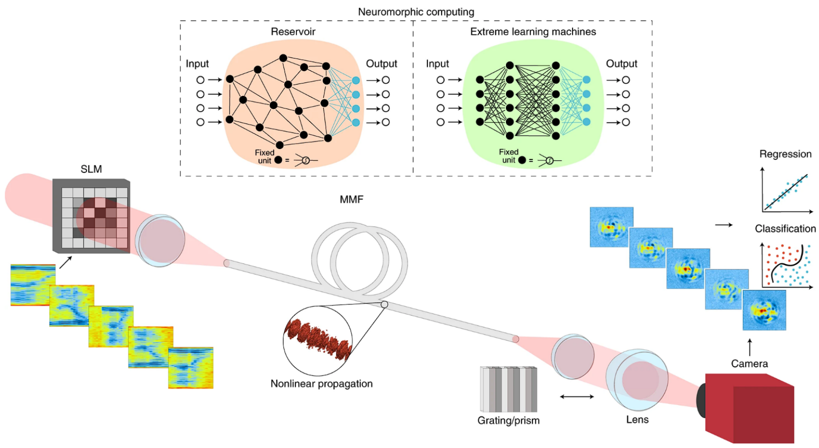

Figure 2.2: illustration of reservoir computing model
[11]

RC can be implemented using physical systems such as multimode fibers (MMFs) with random mode coupling [11] [2]. Here, the randomness and complexity of the physical scattering processes within MMFs serve as the reservoir, enabling efficient computation leveraging the speed of light. The nonlinear transformations and interactions within the MMF act analogously to the reservoir dynamics in traditional RC, enhancing the processing capabilities of optical systems.This enables to use MMFs as optical computing operators.

The key benefits of utilizing physical processes as reservoirs are manifold. Firstly, by harnessing optical non-linearity, computations can be performed at the speed of light, resulting in significantly faster processing times compared to traditional electronic methods. This inherent speed advantage makes optical systems particularly well-suited for applications where rapid data processing is essential. Secondly, optical systems offer improved efficiency by processing large volumes of data while consuming less power than their electronic counterparts.

This energy-efficient characteristic renders optical computing solutions highly desirable for various computational tasks, especially those requiring substantial computational resources. Finally, the randomness and complexity inherent in physical processes contribute to rich reservoir dynamics, enabling effective data transformations without the need for intricate parameter tuning. This inherent resilience to precise parameter adjustment simplifies the implementation of physical reservoirs in practical applications, making them versatile and adaptable solutions for a wide range of computational tasks.[11].

RC implemented with physical reservoirs such as MMFs enables efficient and high-speed computation by leveraging the unique properties of optical systems. This approach holds promise for scalable and energy-efficient solutions to large-scale data processing tasks, particularly in the realm of machine learning and artificial intelligence.

3 Methodology and Implementation

This chapter covers the simulation of the scheme in detailed using python. Section 4.1 explains the methodology behind this implementation scheme. The first parts uses the 'pyMMF' library to generate mode profiles for a Multimode fibre and the second part deals with implementation of Neural Network based on the 'ComplexPyTorch' library to predict a FMF output for a MMF input.

3.1 Few-mode decomposition using multimode reservoir computing

Few-mode decomposition using reservoir computing is an approach to analyze and characterize the modal content of light propagating through few-mode optical fibers (FMFs).

In multimode fiber (MMF) reservoir computing, the input representation consists of intensity measurements captured from the few-mode output, representing complex interference patterns resulting from the superposition of different spatial modes within the multi-mode fiber. The reservoir, a crucial component of the setup, comprises a fixed, randomly initialized network of interconnected neurons.

During the training phase, the reservoir is exposed to input-output pairs derived from known modal content, allowing its dynamics to adjust based on these intensity profiles. This training process aims to capture the underlying relationships between the input (intensity profiles) and the desired output (few-mode decomposition). Subsequently, the reservoir's state after processing the input data is fed into a readout layer. The readout layer, often a trainable linear model such as a simple linear regression or a more complex classifier, learns to map the reservoir's dynamics to the desired output of mode decomposition.

Once trained, the reservoir-computing model is capable of predicting or inferring the modal content of new intensity measurements obtained from FMF outputs, propagating through the MMF and measured after the MMF. By analyzing the reservoir's response to new input data, the model effectively decomposes the input signal into its constituent spatial modes.

3.2 Multi-mode fiber optical simulation

A detailed simulation and analysis of light propagation through a multi-mode optical fiber using the pyMMF library, which is designed for modeling and simulating optical wave-guides. The script is organized into several sections, each contributing to different aspects of the simulation and analysis:

Considering a multi-mode fiber (MMF) with specific parameters such as core radius, numerical aperture (NA), refractive index (n1), and operating wavelength (wl), we initialize the index profile of the fiber using the 'pyMMF' library. This setup enables us to define the spatial mode profiles within the fiber, which are crucial for subsequent mode analysis. By employing a propagation mode solver, we compute the eigenmodes of the MMF, representing distinct spatial patterns through which light propagates within the fiber. These modes are then sorted based on their propagation constants, providing valuable insights into the behavior of light within the MMF. The number of spatial modes generated depends on factors such as the core diameter, with larger diameters resulting in a greater number of modes. This characterization of spatial modes forms the foundation for further analysis and visualization of the MMF's optical properties.

The below figure shows the amplitude and phase of a selected mode from the modes generated for the MMF.

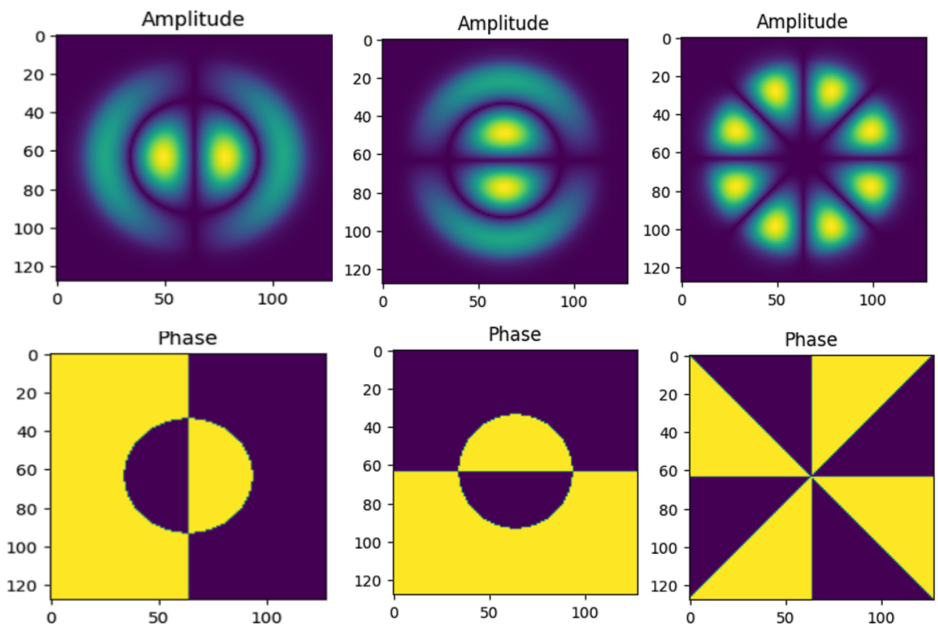

Figure 3.1: Amplitude and Phase of Modes

Random weights ('a' and 'phi') are assigned to each mode to create a complex vector representing a superposition of different fiber modes. These weights are combined with the mode profiles obtained from the solver to simulate the resultant optical field.The intensity distribution within the MMF is calculated based on the squared magnitude of the superposed field ('intensity'), providing insights into the spatial distribution of optical power.

A random complex matrix ('random-matrix') is generated, and Singular Value Decomposition (SVD) is performed using 'np.linalg.svd'. The left singular vectors ('U') obtained from SVD are used as a transformation matrix ('TM') to facilitate mode conversion or transformation between the input and the output of the MMF acting as reservoir.

Multiple random few-mode vectors('fiber-modes-nn') with specified amplitudes and phases are generated. These vectors are normalized and padded to the MMFs dimensionality to match the maximum number of modes ('NmodesMax') for consistency in dimensionality. For each generated few-mode vector, the script calculates the MMF output vector ('y') by applying the transformation matrix ('TM') derived from SVD as in the below equation. It then computes the superposition of modes and visualizes the resulting amplitude and intensity distributions.These intensity distributions are used as the input of the readout layer. The randomly created few-mode vector are used as ground truth.

$$y = TM \times x \; x \;\; = \begin{pmatrix} x_1 & \dots & x_6 & x_0 & \dots & x_0 \end{pmatrix}^\top, \; y = \begin{pmatrix} y_1 \vdots y_6 \; y_7 \vdots y_{55} \end{pmatrix}^\top. \tag{3.1}$$

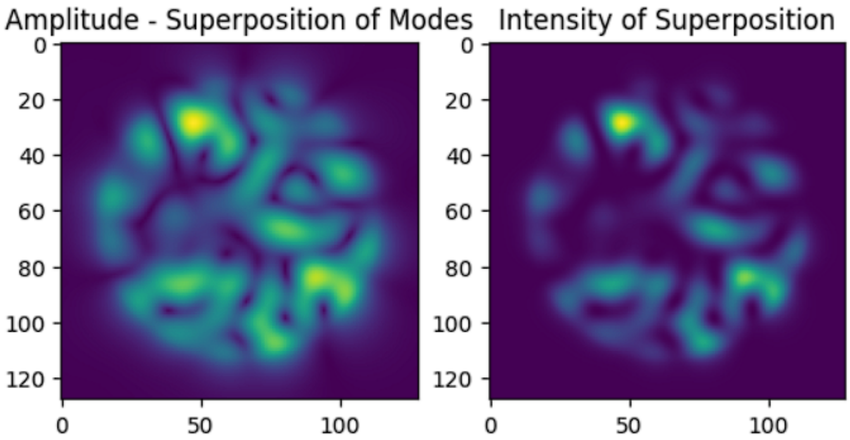

Figure 3.2: Amplitude and Superposition of modes

3.3 Implementation of the reservoir computing readout layer

The "complex PyTorch" repository serves as a specialized extension of the PyTorch deep learning framework tailored specifically for handling complex-valued neural networks. Key features of complex PyTorch include the implementation of complex-valued counterparts to standard PyTorch layers, such as linear layers and convolutional layers, capable of processing both complex-valued inputs and weights. Moreover, complex PyTorch offers support for complex activation functions adapted to work seamlessly with complex inputs,enabling non-linear transformations within complex-valued neural networks.

Now,a deep learning model using the complexPyTorch library is implemented. The goal of this is to train a complex-valued neural network (CNN) for the Implementation of the reservoir computing readout layer.

The 'ComplexNet' class defines a neural network model tailored for handling complex-valued inputs. At its core, the architecture comprises distinct layers designed to process and transform the input data effectively.Starting with the input layer, the network expects complex-valued tensors represented by the variable 'x', with each tensor having dimensions of 128×128.This input layer sets the stage for subsequent processing steps, initializing the flow of data through the network.

After the input layer, a reshape operation is applied to the input tensor. This reshaping is essential to convert the two-dimensional input image into a flattened vector. By reshaping the tensor using 'x.view(-1, 128*128)', the data is prepared for processing by the fully connected layer, ensuring compatibility with the layer's input requirements.

The core of the network lies within the fully connected layer denoted by 'self.fc-single'. This layer performs a linear transformation on the flattened input data, effectively mapping it to the output space. The size of the output space is determined by the 'output-modes' parameter, which specifies the number of output modes or classes the network aims to predict.

Since this is a regression model, an activation function cannot be applied after this fully connected layer as it would introduce non-linearity into the network. Ultimately, the output of the fully connected layer serves as the final output of the network, representing the predicted values or classes corresponding to the input data. Through the sequential processing of input data across its layers, the 'ComplexNet' model enables efficient handling and transformation of complex-valued inputs.

We define helper functions for training ('train') and evaluating ('eval') the complex neural network model. The 'train' function iterates through batches of data, computes the loss using a custom complex-valued loss function ('criterion'), and updates the model parameters using the Adam optimizer ('torch.optim.Adam').The 'eval' function calculates the validation loss using the trained model and specified data loader. During training, progress updates are displayed to monitor training and validation loss. The 'criterion' function defines a custom loss metric for complex-valued data, computing the mean squared error between predicted and target complex numbers. The loss function handles both real and imaginary parts separately to compute a comprehensive loss measure.Additional functions ('correlation-fn','complex-correlation') are defined to calculate custom metrics like correlation coefficients using complex numbers, which can provide insights into the model's performance on complex-valued data.

The main training loop runs for a specified number of epochs ('EPOCHS'), calling the 'train' function to optimize the model parameters based on the training data. The training and validation losses are recorded and plotted using Matplotlib to visualize the training progress over epochs.

4 Results and Analysis

This chapter cover the results achieved after the simulation and analysis of Multimode fibre optical simulation and the implementation results of the reservoir computing readout layer using a Pytorch deep learning framework.

Begin by setting up the parameters for the 55-mode fiber, such as numerical aperture (NA) of 0.1 and core radius of 12.5 microns. Use a propagation mode solver to compute the propagation modes (eigenmodes) of the optical fiber, which represent distinct spatial patterns through which light can propagate within the fiber. Sort the computed modes based on their propagation constants in descending order for further analysis and visualization.

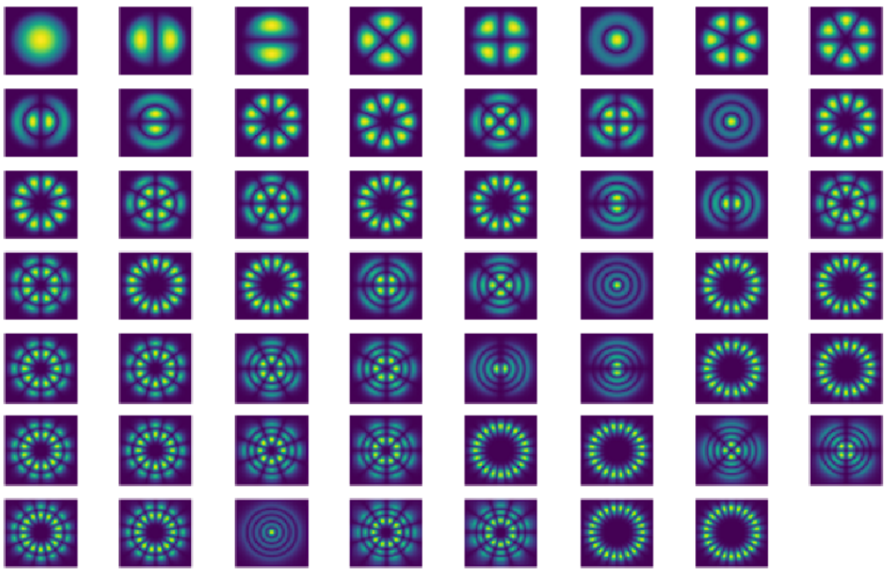

Figure 4.1: Intensity images of the modes from a multimode fiber. The fiber with a core diameter of x and a numerical aperture of y supports 55 orthogonal modes using a wavelength of λ=532nm

Now the few-mode fiber under test is defined. In this work, a FMF with a core radius of x and numerical aperture of 0.1 is used assuming λ = 532nm. This results in 5 orthogonal modes which are aimed to be decomposed. Random data is created using a uniform distribution for both amplitude and phase for each mode of the FMF. This data represents the transmitting FMFs output, which is aimed to be decomposed using the Reservoir-MMF.Apply the transfer matrix (TM) to the generated random data (X) to transform it into a format simulating the scrambling through a 55-mode fiber.

Utilize the mode profiles obtained for the 55-mode fiber to apply superposition with the transformed data (Y). This process creates input data for the neural network (NN) by combining the transformed signals with the mode profiles of the 55-mode fiber.Based on the output of the Reservoir-MMFs field distribution, an intensity-only image is calculated. An example is shown in Fig. 4.2.

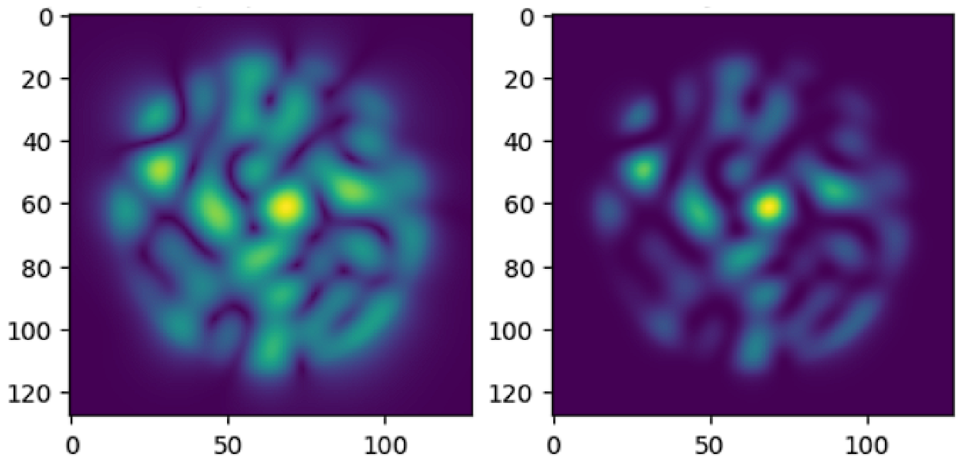

Figure 4.2: Simulation of amplitude (left) and intensity (right) image of a 55-mode Reservoir-MMFs output for a random GT sample of a FMF using 5 modes.

Similarly, superposition is applied with the mode profiles of the 5-mode fiber to the original random data (X) to generate ground truth (GT) signals. These GT signals represent the expected outputs of the NN for comparison during training and evaluation.The figure below generated serves as ground-truth for the reservoir computing based mode decomposition.

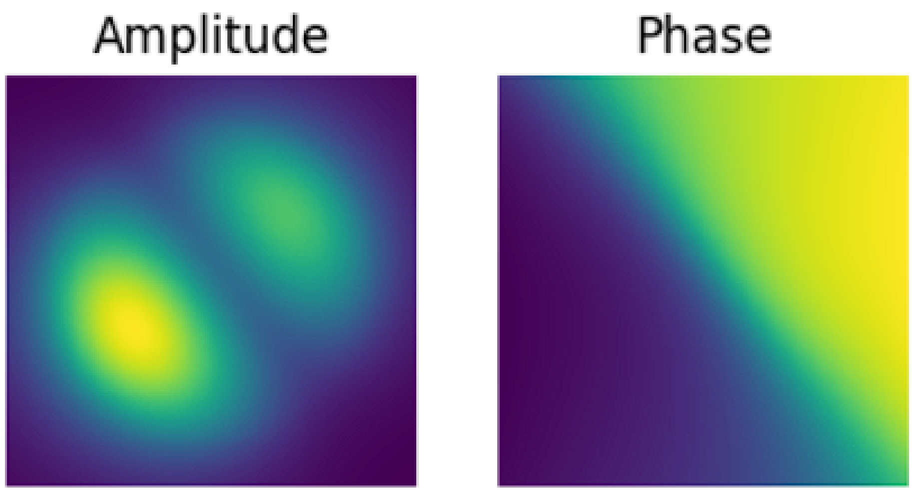

Figure 4.3: Amplitude and Phase images of a randomly created sample from a FMF using 5 modes

First,1000 input samples along with their corresponding ground truths based on the above steps are calculated. This data is split into training, validation and test data by 72:8:20, respectively.Define a neural network model architecture suitable for processing the input data and predicting the corresponding outputs.Train the NN model using the training dataset, optimizing its parameters to minimize the loss function.The readout layer is a single fully connected layer defined as 'self.fc-single = ComplexLinear(128*128, output-modes)'. It's a complex linear layer that takes the flattened input from the previous layer (after flattening the 2D input images into a 1D tensor) and outputs a complex tensor with 'output-modes' number of units.The reservoir size can be inferred from the input size of the fully connected layer in the model. In this case, it's 128x128, which corresponds to the resolution of the images being processed.

The number of trainable parameters in the readout layer can be calculated as the product of the input size and output size of the layer. In this case, it's 128x128x10.The Mean Square Error (MSE) is the loss function for training. The 'criterion' function calculates the MSE between the predicted output and the ground truth.The optimizer used is Adam, with a learning rate of 0.01. This is defined as optimizer = 'torch.optim.Adam(model.parameters(), lr=0.01)'.Monitor and record the training and validation losses over each epoch to track the model's learning progress.The output consists of 5 complex weights in amplitude and phase, corresponding to 'output-modes'.

During training, the readout-layer is optimized whose training process is shown in Fig. 4.4.

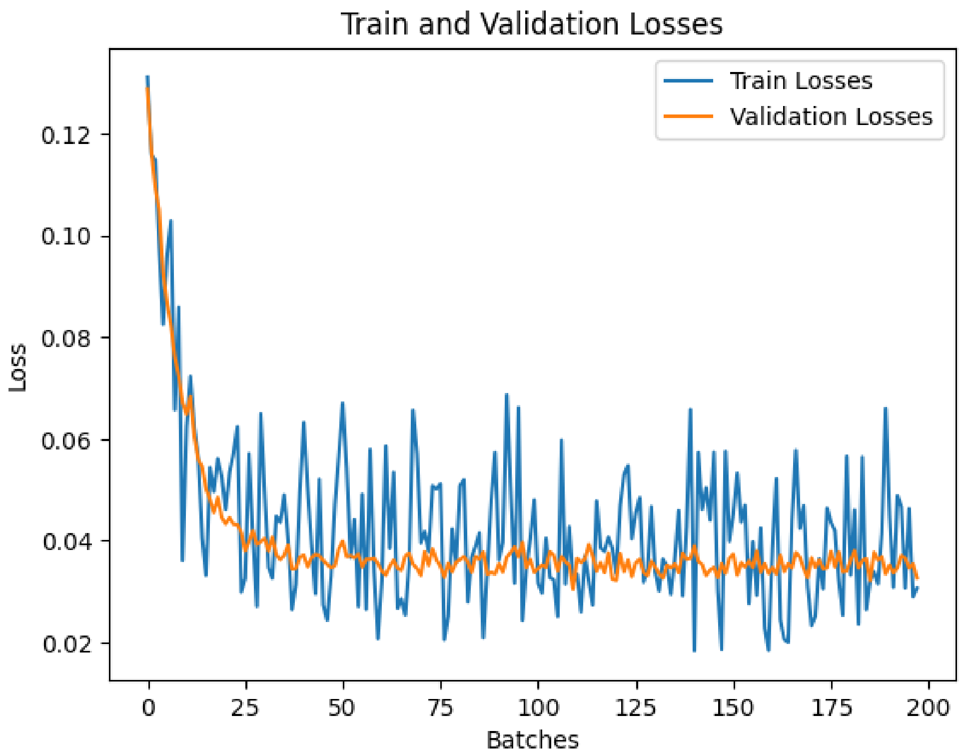

Figure 4.4: Train and Validation Losses Vs No.of Epochs

The trained NN is used to predict the outputs for the testing dataset. The predicted outputs are compared with the ground truth values to assess the performance of the model on unseen data. Visualize the comparison between predicted outputs and ground truths to evaluate the model's accuracy and effectiveness.

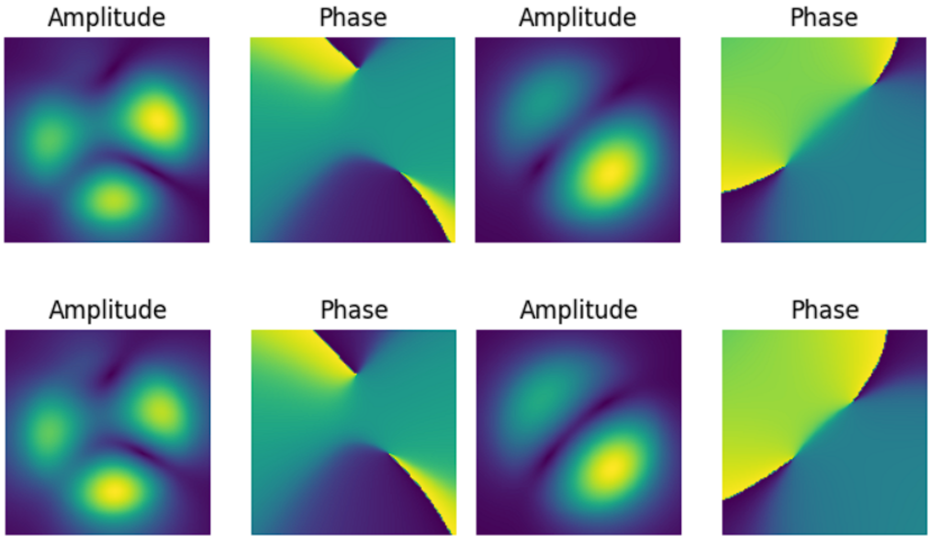

Figure 4.5: Predicted Images (above) Vs Ground Truth images (below)

For all 200 samples of the test data-set, correlation and fidelity is calculated comparing the predicted outputs with the ground truths. The correlation and fidelity values for each image are shown in Figure 4.6 so that you may assess how well the model represents the input-output interactions.200 test samples are randomly computed at the FMF output and then passed through the reservoir-MMF.The mode weights of the FMF are connected to the output camera images of the MMF through a readout layer.

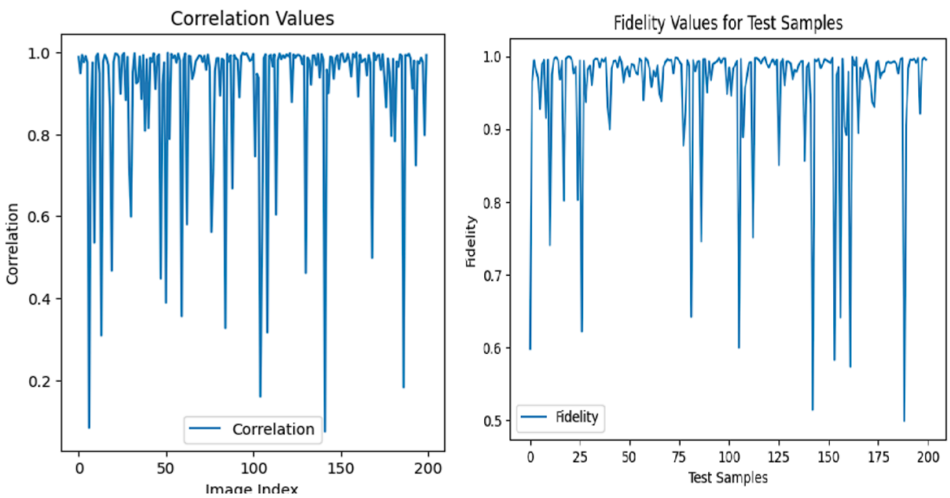

Figure 4.6: Correlation (left) and fidelity (right) for mode decomposition of a 5-mode FMF using a 55-mode Reservoir-MMF

5 Discussion

This chapter discusses the outcomes of the mode decomposition for a 5 mode fiber using a 55 mode fiber operating as a reservoir.

One of the most significant achievements of this research project was the reduction of phase ambiguity within the modal decomposition process using RC. The aim to overcome the challenges associated with phase-based mode decomposition and pave the way for more advanced and effective AI-driven techniques for analyzing multimode fibers has been achieved. Unlike classical neural network models, which struggled with phase ambiguities and trade-offs between prediction quality and response time, the RC approach effectively addressed and reduced phase ambiguities, leading to more accurate and reliable modal decomposition results.

The improvements in machine learning within the past years have gained interest to a bright field of applications. For mode decomposition of a MMF, increasing the number of decomposable channels is a remaining challenge. The main problem here is the scaling of the orthogonal degrees of freedom, which leads to a high complex problem. As for a few-mode fiber a multi-layer perceptron (MLP) could be enough for decomposition of 3 modes, this strongly scales by increasing the number of modes. Already with 5 modes, complex network architectures like VGG or MTNet are necessary for high quality mode decomposition.

In terms of the number of modes decomposed, the RC approach utilizing the 55-mode fiber reservoir successfully decomposed all 5 modes with high accuracy and fidelity. The achieved correlation and fidelity were significantly better compared to the classical neural network models, showcasing the superior performance of RC in modal decomposition tasks.

As within RC the randomness of the reservoir is used, only one readout layer needs to be optimized. By definition, this is a single layer approach achieving a fast computing time. Thus,the RC model demonstrated remarkable efficiency in terms of training parameters, requiring significantly fewer parameters compared to MLP or U-Net. This reduction in training parameters not only streamlined the training process but also contributed to the overall computational efficiency of the model. Another crucial aspect of comparison was the amount of training data required for the modal decomposition task. The RC model exhibited robust performance even with limited training data of one thousand samples, highlighting its ability to generalize well and effectively capture the underlying relationships between input intensity profiles and desired output modal coefficients.

A greater number of modes in the Reservoir-MMF would result in a speckle output with more intricate details and spatial patterns. The increased modal diversity would enrich the speckle output, providing a more comprehensive representation of the input signal's modal content.

With a higher number of modes in the Reservoir-MMF, the computational complexity of the modal decomposition process would increase. The reservoir network would need to accommodate and process a larger number of modes, requiring adjustments in the network architecture and dynamics to effectively capture the modal interactions within the multi-mode fiber.To capture the enhanced details in the speckle output resulting from a higher number of modes in the Reservoir-MMF, a camera with higher resolution and pixel density may be necessary. This would ensure that the camera can accurately capture and analyze the complex speckle patterns generated by the modal decomposition process.

As the number of modes in the Reservoir-MMF grows, the model may require a larger number of trainable parameters to effectively learn and represent the complex modal relationships within the optical signal. However, it is crucial to ensure that the increase in trainable parameters remains significantly lower than that required by traditional models like U-Net to maintain computational efficiency. Despite the potential increase in the number of modes and trainable parameters, the Reservoir-MMF approach should still exhibit superior computational efficiency compared to models like U-Net. By leveraging the speed of light within the multi-mode fiber as an encoder, the Reservoir-MMF can process and analyze optical signals rapidly and efficiently, reducing overall computing effort and enhancing real-time performance.

Environmental conditions can impact the stability of the fiber used in the Reservoir-MMF setup, potentially affecting the modal decomposition performance. Rapid changes in environmental factors such as temperature or humidity may introduce variability in the fiber properties, requiring adjustments in the training process. Reacting quickly with new training data to adapt to these changes is essential to maintain the accuracy and reliability of the modal decomposition results.

Placing a fiber-based Reservoir-MMF at transmission line receivers may pose challenges in terms of practicality and integration. Fiber-based solutions may not be the most suitable for direct integration into transmission line receivers due to space constraints, alignment issues, and potential signal degradation. Considering alternative integration approaches, such as implementing the modal decomposition functionality in integrated photonic devices, could offer more compact and efficient solutions for optical communication systems.

While a camera is necessary for capturing and analyzing the speckle output in modal decomposition, it can be relatively expensive compared to photo-diodes, which are commonly used in state-of-the-art optical communication systems. The cost of integrating a camera into the Reservoir-MMF setup should be carefully considered, especially in large-scale deployment scenarios where cost-effectiveness is a key factor. Exploring cost-efficient alternatives or optimizing camera usage for specific applications can help mitigate this drawback.

6 Summary and Outlook

In summary, the research project focuses on utilizing Reservoir Computing (RC) for multimodal decomposition within optical networks, particularly using multi-mode optical fibers (MMFs). By harnessing the dynamics of a fixed, randomly initialized recurrent neural network (RNN) as the reservoir, the study aims to enhance optical network capabilities through modal decomposition without the need for a second optical path. The RC approach outperforms classical neural network models in terms of accuracy, training efficiency, and generalization capabilities, while also addressing challenges like reducing phase ambiguities. The project highlights the potential of RC as a powerful computational framework for improving optical network performance and efficiency, offering innovative modal decomposition techniques for enhanced optical communication technologies.

The importance of randomness,in cryptographic applications the randomness is essential, especially when creating safe keys that can't be decrypted or accessed by unauthorised parties. The MMF reservoir's intrinsic randomness acts as a natural source of entropy for the generation of encryption keys, guaranteeing strong security protocols for data transferred throughout the network. The reservoir's dynamic behaviour adds an additional degree of unpredictability, which strengthens encryption techniques by using physical layer security. Data security procedures involve encryption keys, which can be generated from the inherent unpredictability in the dynamics of the MMF reservoir. Within the optical network context, encryption keys that are both secure and unique can be created by taking use of the stochastic behaviour of the reservoir. This method provides an innovative means of improving data security and privacy in optical. Within the framework of Reservoir-MMF, the multi-mode fiber's random dynamics can function as a computing resource to carry out specific network operations. By using distributed computing instead of centralised processing, this paradigm can improve network efficiency and open up new functions.

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